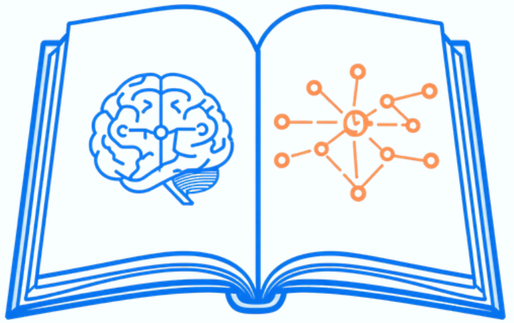


BRAIN TUMOR Classification

Multi-Class MRI Analysis & Comparative Model Benchmarking



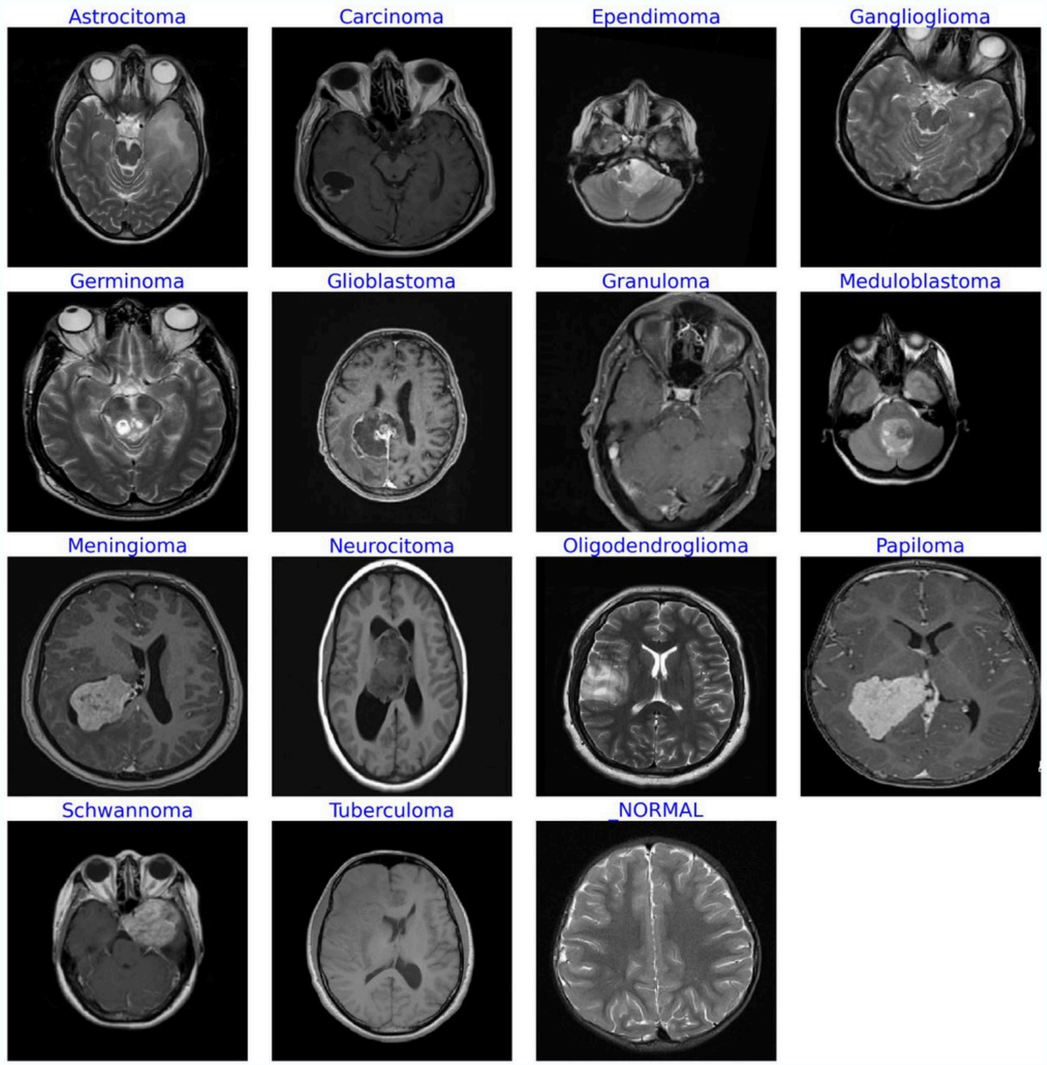
Beatrice Bigoni, Tony Dawra, Dolores Margni

Project Objectives



Develop a multi-class classification system for brain tumors and normal tissue from MRI images, comparing multiple transfer learning architectures to identify the optimal model.

Validate model generalization on an independent external dataset and demonstrate the importance of accurate tumor classification for clinical decision support.

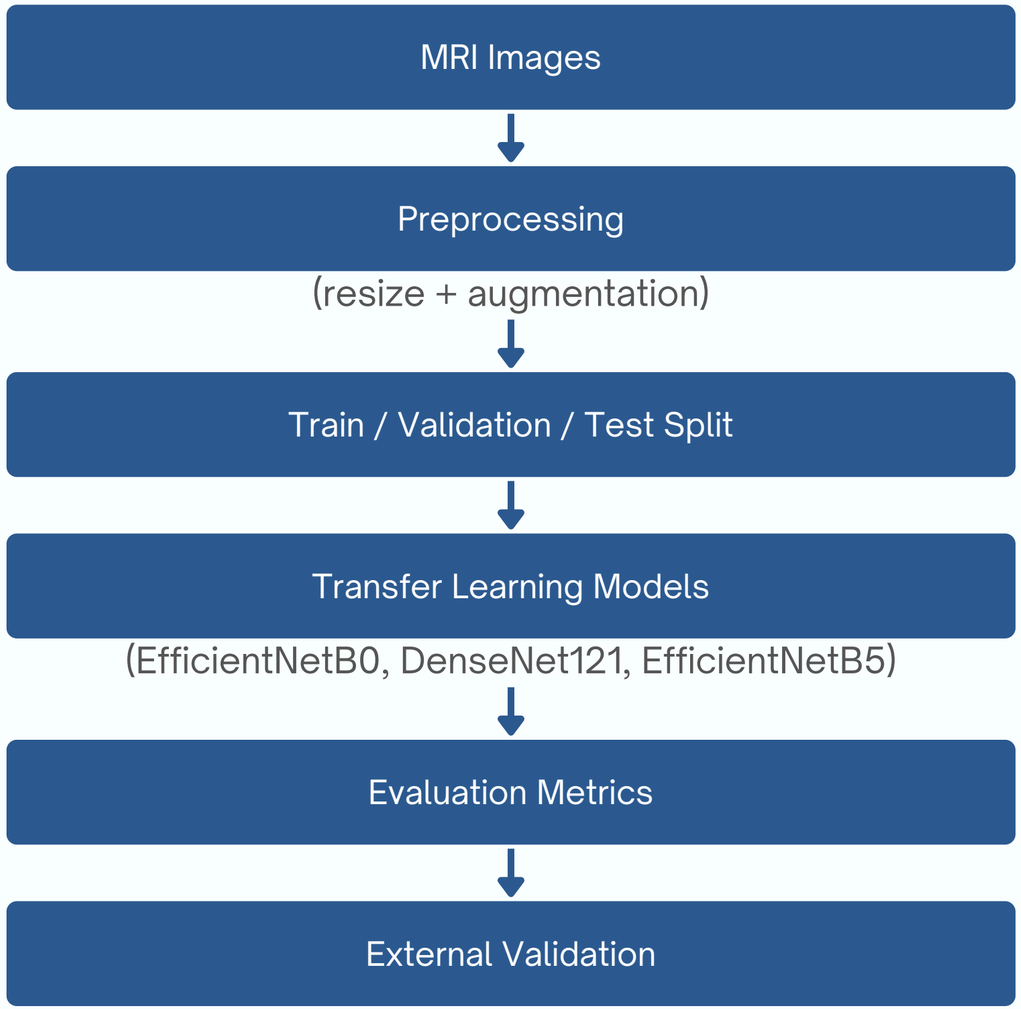


Dataset Overview

- Meningioma
- Astrocitoma
- Glioblastoma
- Schwannoma
- Neurocitoma
- Carcinoma
- Papiloma
- Oligodendroglioma
- Ependimoma
- Tuberculoma
- Meduloblastoma
- Germinoma
- Granuloma
- Ganglioglioma
- NORMAL

Initially, our dataset consisted of 44 classes containing T1, T1C+, and T2 MRI sequences, which we subsequently grouped together.

Methodology Pipeline



Data Preprocessing

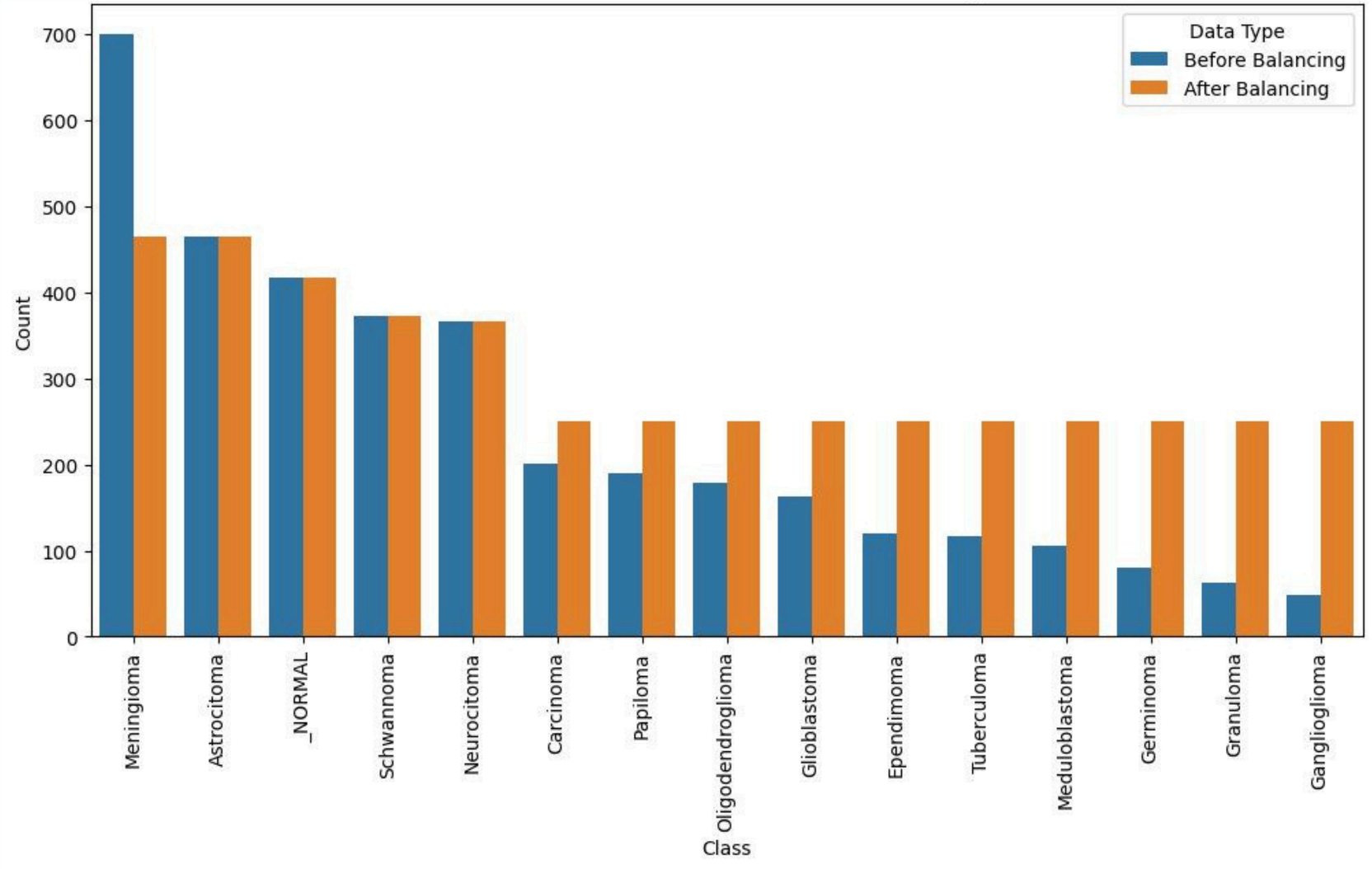


Dataset split: 80% training, 12% validation, 8% test. All images resized to 224x224 pixels for consistent model input dimensions.

Data augmentation applied: rotations, flips, and zooms to balance classes (250-464 samples each) and mitigate class imbalance.

Class Distribution

Before and after balancing



Model 1: EfficientnetB0 (Baseline)

Reliable Efficiency

Architecture: Uses a compound scaling method to balance depth, width and resolution.
Parameter Count: 4.38 Million parameters.
Purpose: Serves as the primary reference

Test Accuracy:
94.13%

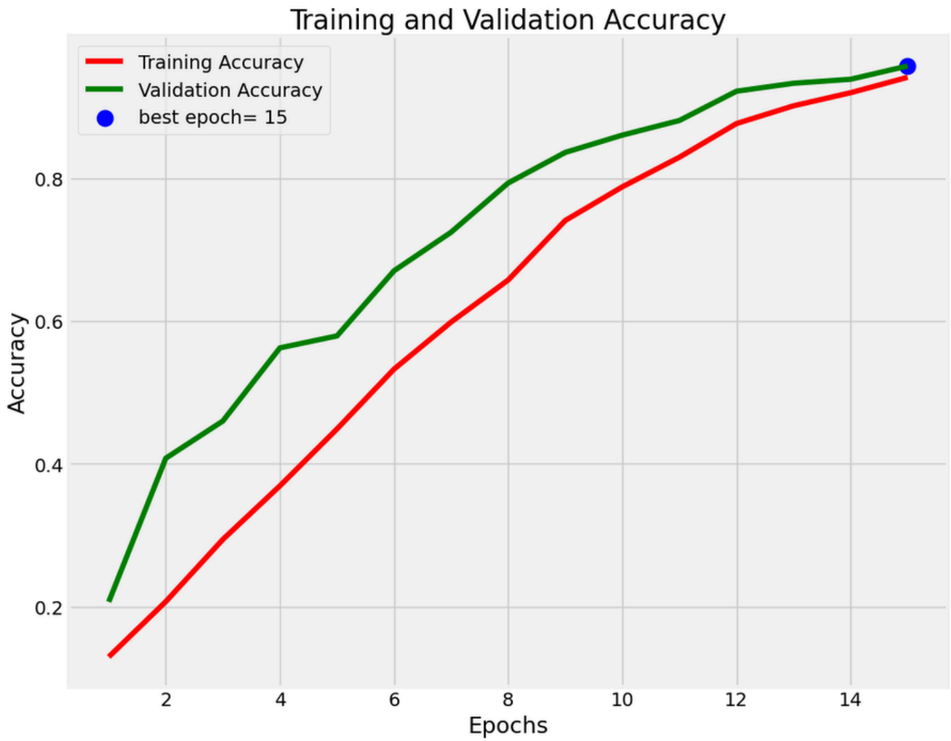
Model: "sequential"

Layer (type)	Output Shape	Param #
efficientnetb0 (Functional)	(None, 1280)	4,049,571
flatten (Flatten)	(None, 1280)	0
dropout (Dropout)	(None, 1280)	0
dense (Dense)	(None, 256)	327,936
dropout_1 (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 15)	3,855

Total params: 4,381,362 (16.71 MB)
Trainable params: 4,339,339 (16.55 MB)
Non-trainable params: 42,023 (164.16 KB)

Model 1: EfficientnetB0 (Baseline)





Model 1b: EfficientnetB0 (Simplified)

Compressed Head

Modification: Reduced complexity in the top classification layers.
Optimization: Lower inference time but significant drop in feature discrimination.
Finding: A simplified head is insufficient for a complex 15-class problem.

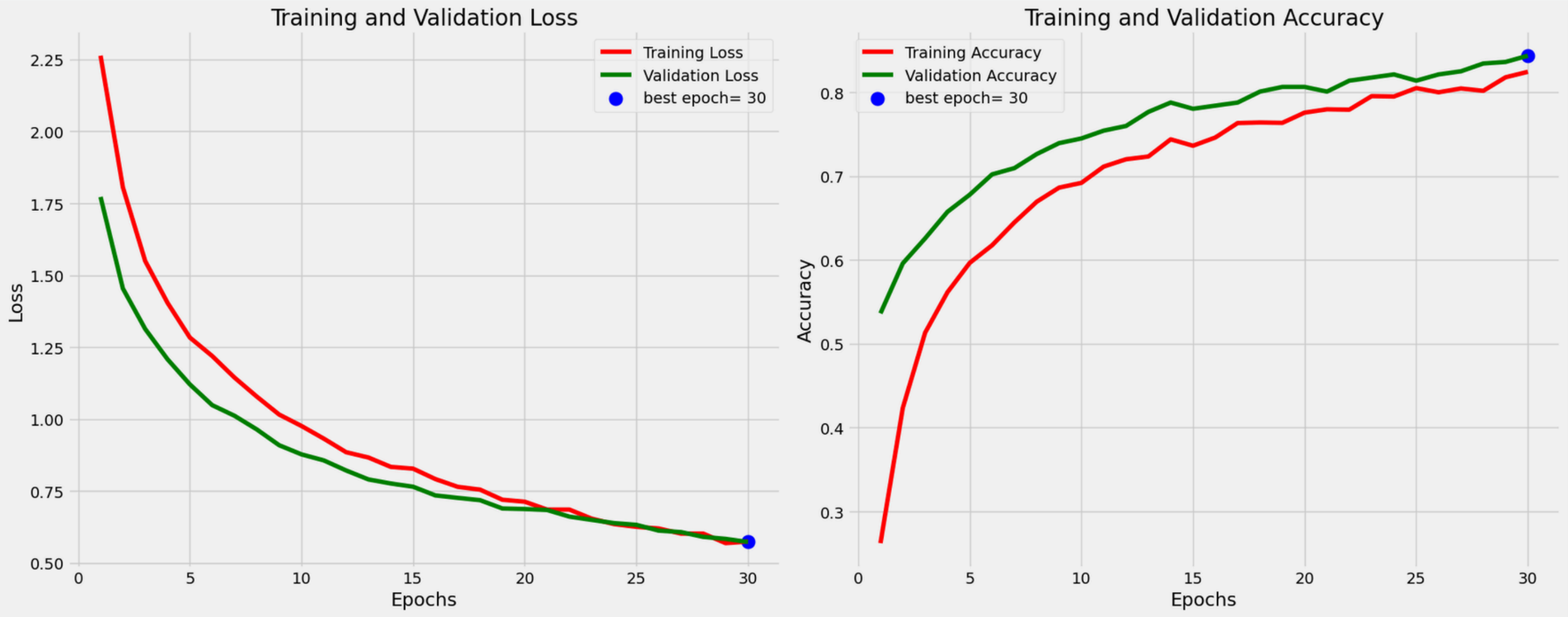
Test Accuracy:
83.52%

Model: "sequential_1"

Layer (type)	Output Shape	Param #
efficientnetb0 (Functional)	(None, 1280)	4,049,571
dropout_2 (Dropout)	(None, 1280)	0
dense_2 (Dense)	(None, 128)	163,968
dropout_3 (Dropout)	(None, 128)	0
dense_3 (Dense)	(None, 15)	1,935

Total params: 4,215,474 (16.08 MB)
Trainable params: 165,903 (648.06 KB)
Non-trainable params: 4,049,571 (15.45 MB)

Model 1b: EfficientnetB0 (Simplified)



Model 2: EfficientNetB5

Maximum Feature Extraction

Superiority: Significant increase in depth and width allows for detecting micro-anatomical tumor variations.
Multi-class Expert: Best performance in distinguishing highly similar tumor tissues.
Complexity: 29.04 Million parameters (High resource demand).

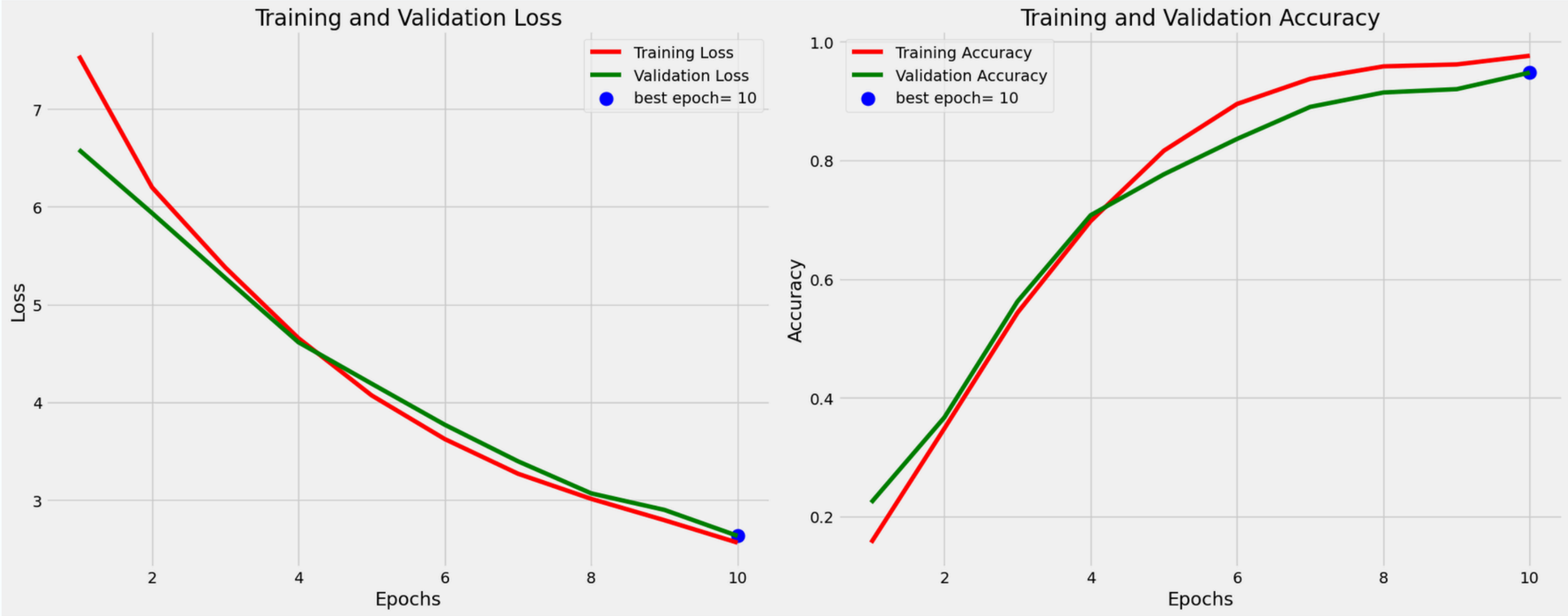
Test Accuracy:
94.41%

Model: "sequential_2"

Layer (type)	Output Shape	Param #
efficientnetb5 (Functional)	(None, 2048)	28,513,527
flatten_1 (Flatten)	(None, 2048)	0
dropout_4 (Dropout)	(None, 2048)	0
dense_4 (Dense)	(None, 256)	524,544
dropout_5 (Dropout)	(None, 256)	0
dense_5 (Dense)	(None, 15)	3,855

Total params: 29,041,926 (110.79 MB)
Trainable params: 28,869,183 (110.13 MB)
Non-trainable params: 172,743 (674.78 KB)

Model 2: EfficientNetB5



Model 3: Densenet121

Feature Connectivity

Dense Connections: Each layer is connected to every other layer, maximizing information flow.
Feature Reuse: Excellent at maintaining small details throughout the network.
Parameters: 7.30 Million (Moderate complexity).

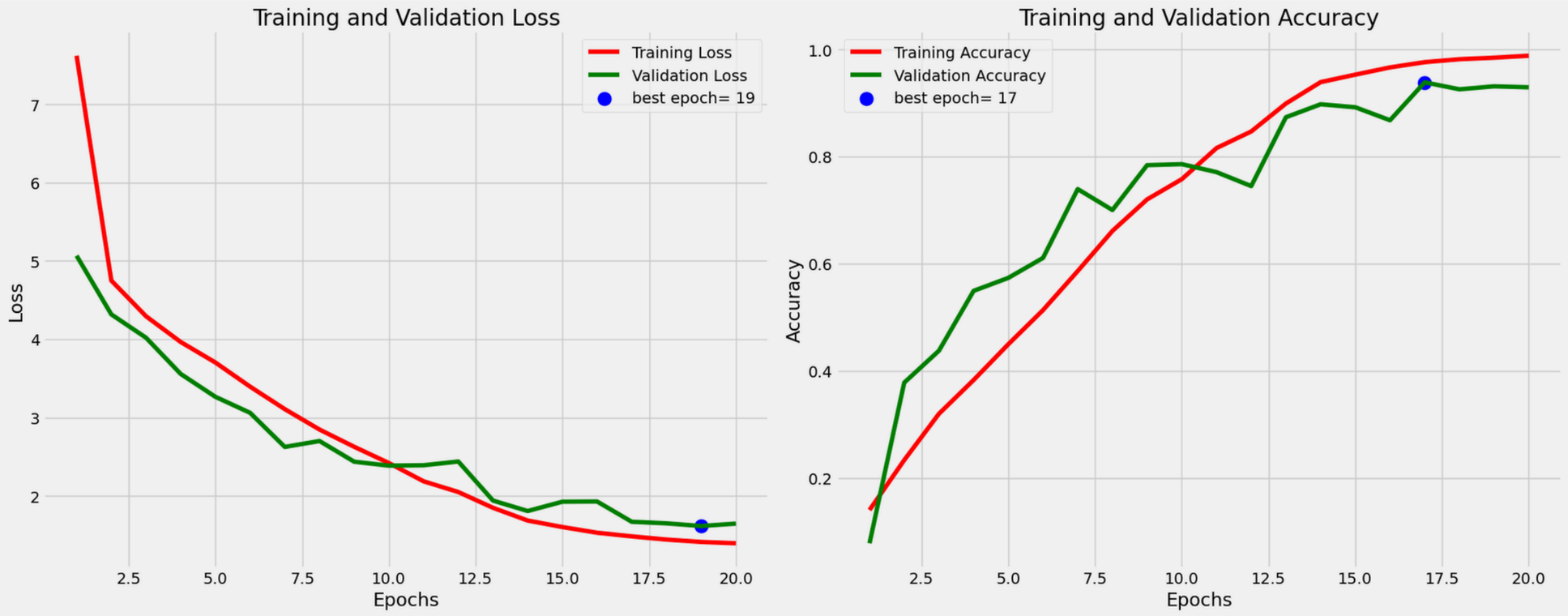
Test Accuracy:
93.02%

Model: "sequential_3"

Layer (type)	Output Shape	Param #
densenet121 (Functional)	(None, 1024)	7,037,504
flatten_2 (Flatten)	(None, 1024)	0
dropout_6 (Dropout)	(None, 1024)	0
dense_6 (Dense)	(None, 256)	262,400
dropout_7 (Dropout)	(None, 256)	0
dense_7 (Dense)	(None, 15)	3,855

Total params: 7,303,759 (27.86 MB)
Trainable params: 7,220,111 (27.54 MB)
Non-trainable params: 83,648 (326.75 KB)

Model 3: Densenet121



Model 4: Custom Model

Feature Connectivity

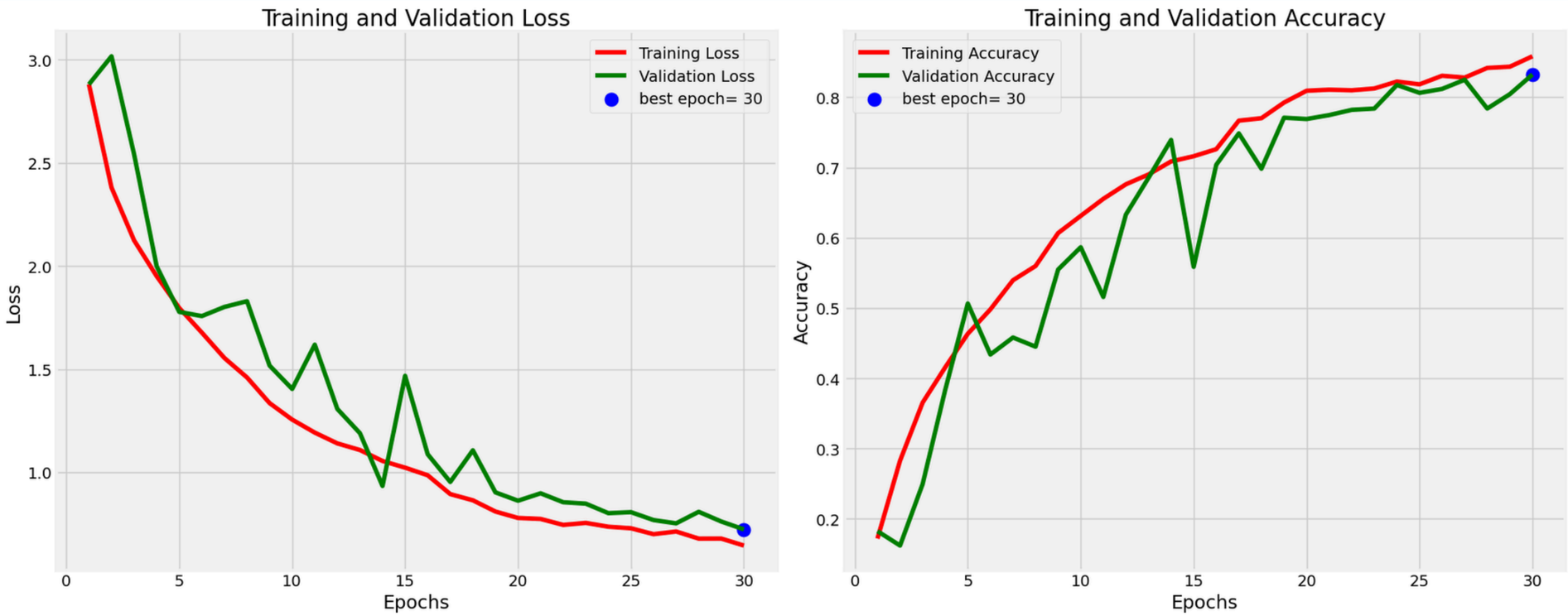
Deep Convolutions: Four convolutional blocks progressively extract increasingly complex MRI features.
Regularization: Batch Normalization and Dropout improve training stability and reduce overfitting.
Parameters: 1.21 Million.

Test Accuracy:
75.14%

Network Section	Layer Types / Operations	Output Shape
Input	Image Input	(224, 224, 3)
Conv Block 1	2x (Conv2D 32 $\rightarrow$ BN $\rightarrow$ ReLU) + MaxPool + Dropout	(112, 112, 32)
Conv Block 2	2x (Conv2D 64 $\rightarrow$ BN $\rightarrow$ ReLU) + MaxPool + Dropout	(56, 56, 64)
Conv Block 3	2x (Conv2D 128 $\rightarrow$ BN $\rightarrow$ ReLU) + MaxPool + Dropout	(28, 28, 128)
Conv Block 4	2x (Conv2D 256 $\rightarrow$ BN $\rightarrow$ ReLU) + MaxPool + Dropout	(14, 14, 256)
Classifier Head	GlobalAvgPool2D -> Dense (128) -> BN -> ReLU -> Dropout | Shape: (128)	(128)
Output Layer	Dense (Softmax) | Shape: (15)	(15)

Total params: 1,210,351 (~1.21 M)
Trainable Params: 1,208,175
Non-Trainable params: 2176
Key Features: Deep convolutional architecture built from scratch. Utilizes Batch Normalization and Dropout layers to improve training stability and prevent overfitting.

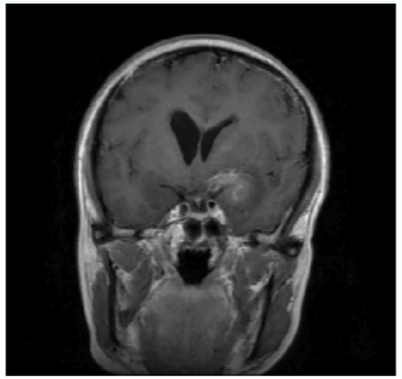
Model 4: Custom Model



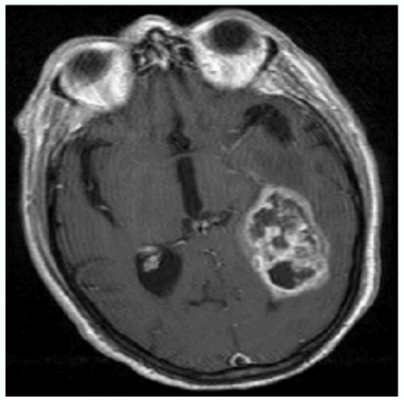
Models Comparison

Model	Parameters	Train Accuracy	Validation Accuracy	Test Accuracy	Macro F1
EfficientNetB0_Original	4.38 M	94.13%	95.71%	94.13%	0.9140
EfficientNetB0_Simplified	4.21 M	82.43%	84.35%	83.51%	0.8133
EfficientNetB5	29 M	97.62%	94.78%	94.41%	0.9198
DenseNet121	7.30 M	98.82%	93.85%	93.01%	0.9129
Custom CNN	1.21 M	85.83%	83.24%	75.13%	0.7281

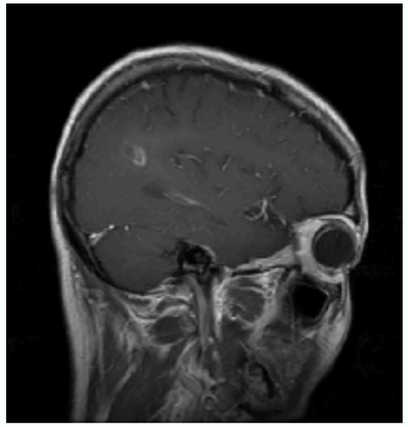
Coronal view



Axial view



Sagittal view



External validation

Model	Accuracy	F1-Score
EfficientNetB0_Simplified	41.83%	0.3891
EfficientNetB0_Original	44.25%	0.4047
EfficientNetB5	47.42%	0.4402
DenseNet121	37.92%	0.3042
Custom CNN	37.08%	0.3204

↓

Domain shift problem

Second External validation
Part 1

```
Glioma (Astrocytoma, Ganglioglioma, Glioblastoma, Oligodendroglioma, Ependymoma)
Meningioma (Low Grade, Atypical, Anaplastic, Transitional)
Neurocytoma (Central - Intraventricular, Extraventricular)
NORMAL
Other Types of Injuries (Abscesses, Cysts, Miscellaneous Encephalopathies)
Schwannoma (Acoustic, Vestibular - Trigeminal)
```

```
Matched classes used in evaluation:
Astrocitoma: class id 0
Meningioma: class id 8
Neurocitoma: class id 9
Schwannoma: class id 12
_NORMAL: class id 14
```

Model	Accuracy	F1-score
B0-Simplified	0.7761	0.7988
B0-Baseline	0.8271	0.8431
B5	0.8110	0.8383
Densenet	0.8516	0.8708
Custom	0.7667	0.7866

Second External validation
part 2

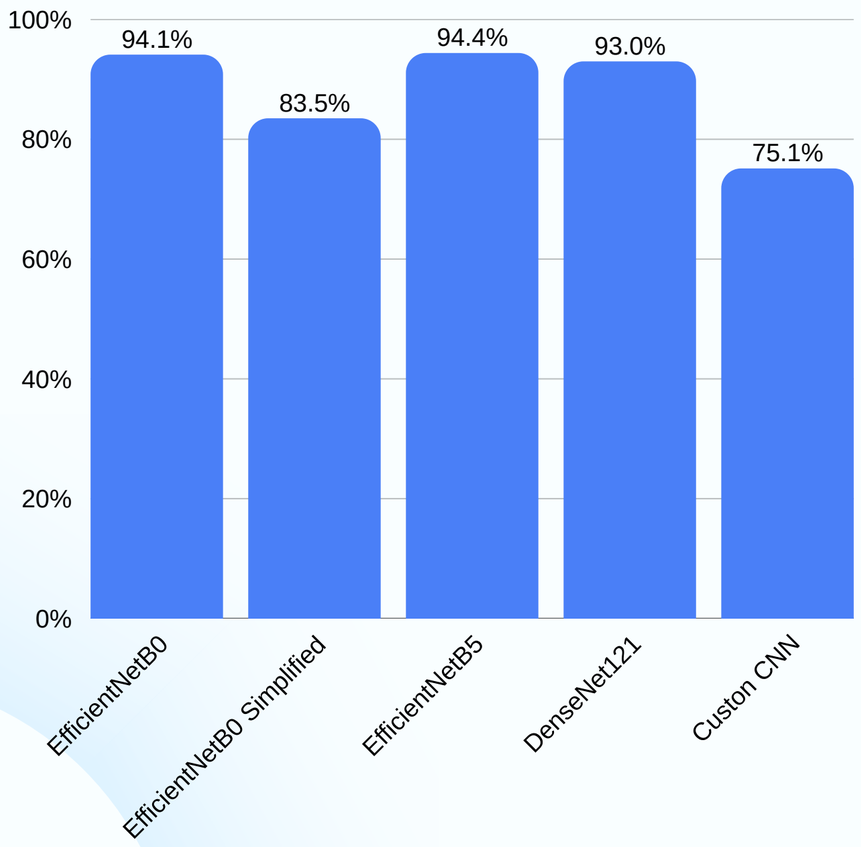
Model	Accuracy	F1-score
B0-Simplified	0.8218	0.7057
B0-Baseline	0.8718	0.7477
B5	0.8790	0.7542
Densenet	0.8949	0.7659
Custom	0.7494	0.6529

```
Grouped classes used:
['Glioma_Group', 'Meningioma', 'Neurocitoma', 'Other_Lesions', 'Schwannoma', '_NORMAL']

Superclass to original-class indices:
Glioma_Group -> [0, 2, 3, 5, 10]
Meningioma -> [8]
Neurocitoma -> [9]
Other_Lesions -> [1, 4, 6, 7, 11, 13]
Schwannoma -> [12]
_NORMAL -> [14]
```

```
Glioma (Astrocytoma, Ganglioglioma, Glioblastoma, Oligodendroglioma, Ependymoma)
Meningioma (Low Grade, Atypical, Anaplastic, Transitional)
Neurocytoma (Central - Intraventricular, Extraventricular)
NORMAL
Other Types of Injuries (Abscesses, Cysts, Miscellaneous Encephalopathies)
Schwannoma (Acoustic, Vestibular - Trigeminal)
```

Key Results



In the internal test, the best model is EfficientNetB5 (94.4%), closely followed by EfficientNetB0 (94.1%) and DenseNet121 (93.0%). However, in external validation, DenseNet121 generalizes best. A possible reason is model complexity: DenseNet121 has far fewer parameters than EfficientNetB5, so it may be less prone to learning dataset-specific patterns and more robust under domain shift. This shows that the best internal performance does not always guarantee the best external robustness.

Alternative Experiments & Lessons Learned

1. Data balancing
Full over/undersampling toward the average class size reduced performance, so a softer strategy was used: slight oversampling of minority classes and limited undersampling of majority classes.

2. Model selection
We started with EfficientNetB5 and DenseNet121, then added EfficientNetB0 and B0-Simplified to compare larger models with lighter pretrained architectures.

3. Custom CNN experiment
A from-scratch Custom CNN was added as a baseline. The first version showed clear overfitting.

Alternative Experiments & Lessons Learned

4. Custom CNN improvement
After applying regularization, the Custom CNN improved and reached 75% test accuracy, but transfer learning models still performed better.

5. Freezing vs fine-tuning
B0-Simplified with a frozen backbone reached 83.5%, while full EfficientNetB0 reached 94.1%, showing that unfreezing helped MRI-specific adaptation.

6. External validation
The first external test performed poorly due to MRI orientation/domain shift. A second external test used matched/grouped labels, with unavailable classes grouped under Other_Lesions.

Conclusion

This project showed that transfer learning is effective for brain tumor MRI classification. EfficientNetB5 achieved the best internal accuracy, while DenseNet121 generalized best on external data.

The Custom CNN improved after regularization but remained weaker than pretrained models. External testing also showed that MRI orientation and label differences can affect performance.

In real life, this type of model can support doctors in brain tumor classification, but it should not replace medical experts. Final diagnosis must always rely on doctors and clinical validation.

